QF620 Stochastic Modelling in Finance — Session 1 Reference Cheatsheet

Probability Theory and Lattice Models: expectation, moments and central moments, skewness and kurtosis, moment generating functions, normal and lognormal variables, two-state and three-state models

Built from Session 1 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Session 1: Probability Theory and Lattice Models" (Tee Chyng Wen, SMU, 22 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. From this

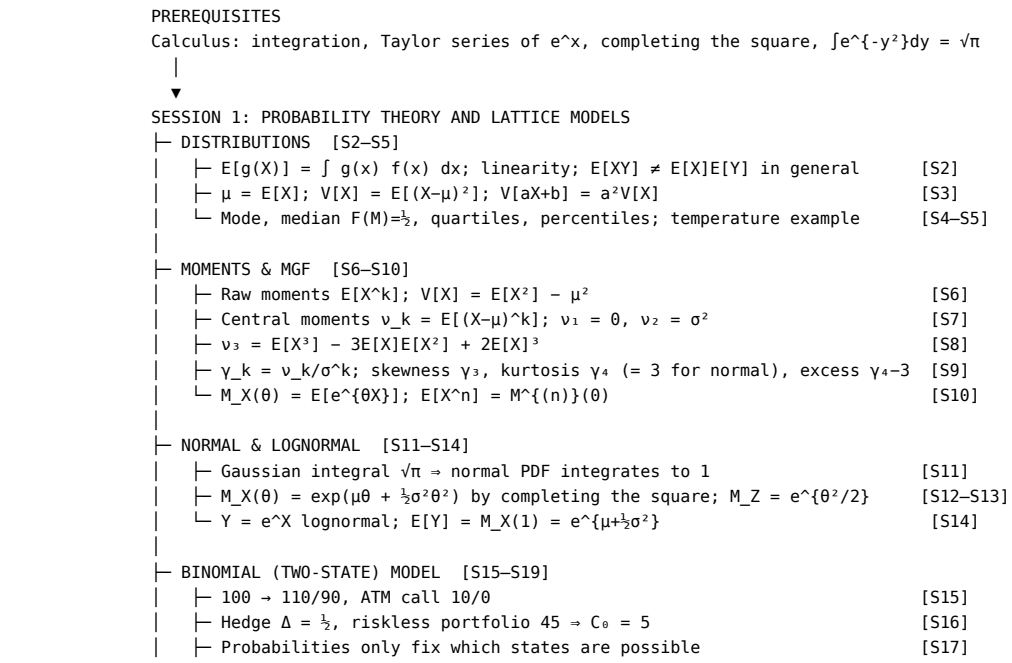
session you need: expectation and variance rules; moments vs central moments and expressing one in terms of the other; skewness and (excess) kurtosis; using the MGF to get moments; the normal MGF (and its derivation by completing the square); the lognormal mean; and pricing by hedging and by risk-neutral probabilities in the two-state and three-state models.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
X	random variable	random	
$f(x)$	probability density function (PDF)	function	$\int f = 1,\ f \geq 0$ [S2]
$F(x)$	cumulative distribution function (CDF)	function	$F(x) = \mathbb{P}(X \leq x)$ [S4]
$\mathbb{E}[\cdot]$	expectation	number	[S2]
$V[\cdot]$	variance	number	also $\mathbb{V}$ elsewhere in the course [S3]
μ	mean $\mathbb{E}[X]$	number	[S3]
$\sigma^2,\ \sigma$	variance; standard deviation	number	[S3]
$\mathbb{E}[X^k]$	k-th (raw) moment	number	about the origin [S6]
ν_k	k-th central moment $\mathbb{E}[(X-\mu)^k]$	number	about the mean [S7]
γ_k	standardised moment ν_k/σ^k	number	γ_3 skewness, γ_4 kurtosis [S9]
$M_X(\theta)$	moment generating function	function	$\mathbb{E}[e^{\theta X}]$ [S10]
$N(\mu, \sigma^2)$	normal distribution	distribution	[S11–S13]
Z	standard normal $N(0,1)$	random	[S13]
$\phi(x)$	standard normal PDF	function	[S11]
$Y = e^X$	lognormal variable	random	[S14]
$S_0,\ S_1$	stock today / tomorrow	price	[S15]
$C_0,\ C_1$	call value today / tomorrow	price	[S15]
K	strike	price	
Δ	number of shares held (hedge ratio)	number	[S16]
$p,\ p^*$	real-world / risk-neutral up probability	probability	[S18–S19]

Notation trap: the slides write variance as $V[X]$ in Sessions 1 and 4 and the kurtosis numerator once as v_4 (a typo for ν_4) [S9]. σ is the standard deviation of X here; from Session 2 on it is a volatility per $\sqrt{\text{time}}$.

0. Topic map



```
|   ├─ Risk-neutral p* = ½ reproduces C₀ = 5 and E[S₁] = S₀                  [S18]
|   └─ No risk premium needed: hedging removes it                            [S19]
|
└─ TRINOMIAL (THREE-STATE) MODEL  [S20–S22]
    ├─ Add a middle state 100; cannot hedge 3 states with 1 Δ                [S20–S21]
    ├─ Hedging gives only bounds: 0 < C₀ < 5                                 [S21]
    └─ p* ∈ (0, ½) ⇒ price interval (0,5): incomplete market                 [S22]
  |
  ▼
LATER: S2 multi-period binomial trees, Radon–Nikodym, martingales; S3 Brownian motion
```

Dependency logic:

- Expectation is the basic tool; moments, variance, skewness, kurtosis and the MGF are all expectations of particular functions $g(X)$.
- The MGF packs every moment into one function. For the normal it has a closed form, which immediately gives the lognormal mean $e^{\mu+\frac{1}{2}\sigma^2}$, the convexity term behind $r-\frac{1}{2}\sigma^2$ in Session 2.
- The two-state model shows that a hedge, not a probability, fixes the option price; the risk-neutral p^* is just a device that reproduces that price. With three states the hedge fails and only a price interval survives.

1 Concept entries

1.1 Expectation and its properties [S2]

The PDF $f(x)$ contains all information about X. The expectation of a function of X:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, f(x)\, \mathrm{d}x.$$

where

X	a continuous random variable.
$g(X)$	any function of X (e.g. X, X^2, $e^{\theta X}$, an option payoff).
$f(x)$	probability density function of X: $f(x)\,\mathrm{d}x \approx \mathbb{P}(x < X \le x + \mathrm{d}x)$.
$\int_{-\infty}^{\infty} \ldots \mathrm{d}x$	probability-weighted average of g over every possible value x. For a discrete X replace with $\sum_x g(x)\mathbb{P}(X=x)$.

Properties [S2]:

$$\mathbb{E}[a] = a, \quad \mathbb{E}[a\,g(X)] = a\,\mathbb{E}[g(X)], \quad \mathbb{E}[s(X)+t(X)] = \mathbb{E}[s(X)] + \mathbb{E}[t(X)], \quad \mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y] \text{ in general.}$$

where

a	a constant (non-random).
$s(X),\ t(X)$	two functions of X; expectation is **linear**.
$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$	holds when X and Y are *independent* (more generally, uncorrelated); otherwise $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \mathrm{Cov}(X,Y)$ *[Beyond slides].*

1.2 Mean and variance [S3]

$$\mu = \mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x, \qquad V[X] = \mathbb{E}\big[(X-\mu)^2\big] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, \mathrm{d}x.$$

where

μ	mean (first moment): the centre of mass of the distribution.
$x\, f(x)$	each value weighted by its density.
$(X-\mu)^2$	squared distance of X from its mean.
$V[X] = \sigma^2$	variance: average squared distance from the mean, measuring spread.

Properties [S3]:

$$V[a] = 0, \qquad V[aX+b] = a^2\, V[X].$$

where

$a,\ b$	constants.
$V[a] = 0$	a constant does not vary.
a^2	scaling X by a scales distances by a and squared distances by a^2.
b	a shift moves the mean but not the spread, so it drops out.

- **Standard deviation** $\sigma = \sqrt{V[X]}$: spread in the same units as X.
- △**!** [S3] says "the variance is always positive". It is always *non-negative*: $V[a] = 0$ for a constant, as the slide itself states.

1.3 Mode, median, quartiles and percentiles [S4–S5]

Measures of the "average" that need no expectation. **Mode**: value of X where $f(x)$ is largest (several maxima ⇒ several modes). **Median** M, quartiles and percentiles [S4]:

$$F(Q_{\text{lower}}) = \tfrac{1}{4}, \qquad F(M) = \tfrac{1}{2}, \qquad F(Q_{\text{upper}}) = \tfrac{3}{4}, \qquad F(P_n) = \tfrac{n}{100}.$$

where

$F(x)$	cumulative distribution function $\mathbb{P}(X \le x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u$.
M	median: half the probability lies below it.
Q_{lower}, Q_{upper}	lower and upper quartiles: a quarter / three quarters of the probability lies below them. Together with M they cut the distribution into 4 equal-probability regions.
P_n	n-th percentile: n% of the probability lies below it.

Slide example [S5] (worked here, *[Beyond slides]*): 110 days; days 1–100 at 80°, 101–105 at 85°, 106–110 at 90°.

- Sorted, the 55th and 56th observations are both 80, so the **median is 80°**. The mode is also 80°.
- Mean $= (100 \cdot 80 + 5 \cdot 85 + 5 \cdot 90)/110 = 8875/110 \approx 80.68°$: the few hot days pull the mean up but leave the median unchanged. The median is robust to outliers.

1.4 Moments and central moments [S6–S8]

Raw moments [S6]:

$$\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f(x)\,\mathrm{d}x, \qquad V[X] = \mathbb{E}[X^2] - \mu^2.$$

where

k	order (power) of the moment: $k = 1$ is the mean.
$\mathbb{E}[X^k]$	k-th moment "about the origin" $x = 0$.
$\mathbb{E}[X^2] - \mu^2$	variance from the first two raw moments (expand $\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2$).

Central moments [S7]: moments about the mean.

$$\nu_k = \mathbb{E}\big[(X-\mu)^k\big] = \int_{-\infty}^{\infty} (x-\mu)^k f(x)\,\mathrm{d}x, \qquad \nu_1 = 0, \quad \nu_2 = V[X] = \sigma^2.$$

where

ν_k	k-th central moment (Greek "nu").
$X - \mu$	deviation from the mean; "central" because we subtract μ before raising to the power.
$\nu_1 = 0$	deviations average to zero.
ν_2	the variance (second central moment).

Slide example [S8]: third central moment in terms of raw moments.

$$\nu_3 = \mathbb{E}\big[(X-\mu)^3\big] = \mathbb{E}[X^3] - 3\mathbb{E}[X]\mathbb{E}[X^2] + 2\mathbb{E}[X]^3.$$

where

$(X-\mu)^3$	expanded: $X^3 - 3\mu X^2 + 3\mu^2 X - \mu^3$.
$\mathbb{E}[X^3]$	third raw moment.
$-3\mathbb{E}[X]\mathbb{E}[X^2]$	from $-3\mu\mathbb{E}[X^2]$ with $\mu = \mathbb{E}[X]$.
$2\mathbb{E}[X]^3$	$3\mu^2 \cdot \mu - \mu^3 = 2\mu^3$.

[Beyond slides] Likewise $\nu_4 = \mathbb{E}[X^4] - 4\mu\mathbb{E}[X^3] + 6\mu^2\mathbb{E}[X^2] - 3\mu^4$.

1.5 Skewness and kurtosis [S9]

$$\gamma_k = \frac{\nu_k}{\sigma^k}, \qquad \gamma_3 = \text{skewness}, \qquad \gamma_4 = \text{kurtosis}, \qquad \gamma_4^{N(\mu,\sigma^2)} = \frac{\nu_4}{\sigma^4} = \frac{3\sigma^4}{\sigma^4} = 3.$$

where

γ_k	standardised (normalised, dimensionless) central moment.
σ^k	standard deviation to the k-th power; dividing removes the units of X.
γ_3	skewness: 0 if symmetric; < 0 (negatively skewed) if the long tail is on the left (values below the mean); > 0 if it is on the right.
γ_4	kurtosis: weight in the tails relative to the shoulders.
$3\sigma^4$	fourth central moment of a normal distribution.

Excess kurtosis $= \gamma_4 - 3$. Positive excess kurtosis: narrower peak and fatter tails than a normal with the same mean and variance (typical of asset returns *[Beyond slides]*).

- △**!** [S9] writes the normal kurtosis as v_4/σ^4; it is ν_4/σ^4 (Greek nu).

1.6 Moment generating function [S10]

$$M_X(\theta) = \mathbb{E}\left[e^{\theta X}\right] = \int_{-\infty}^{\infty} e^{\theta x} f(x)\,\mathrm{d}x, \qquad M_X(0) = 1.$$

where

θ	a dummy real variable (not a parameter of the distribution).
$e^{\theta X}$	the random variable whose expectation generates the moments.
$M_X(0) = 1$	because $e^0 = 1$ and $\int f = 1$.

Expanding the exponential as a Taylor series:

$$\mathbb{E}\left[e^{\theta X}\right] = \mathbb{E}\left[1 + \theta X + \frac{\theta^2 X^2}{2!} + \frac{\theta^3 X^3}{3!} + \cdots\right] = 1 + \mathbb{E}[X]\,\theta + \mathbb{E}[X^2]\frac{\theta^2}{2!} + \mathbb{E}[X^3]\frac{\theta^3}{3!} + \cdots$$

where

$\frac{\theta^n X^n}{n!}$	n-th term of $e^y = \sum y^n/n!$ with $y = \theta X$.
$\mathbb{E}[X^n]$	the n-th moment appears as the coefficient of $\theta^n/n!$ (linearity of $\mathbb{E}$).

So the moments come from derivatives at zero:

$$\mathbb{E}[X^n] = \left.\frac{\mathrm{d}^n M_X(\theta)}{\mathrm{d}\theta^n}\right|_{\theta=0}.$$

where

$\frac{\mathrm{d}^n}{\mathrm{d}\theta^n}$	n-th derivative with respect to θ; it kills all lower powers and leaves $\mathbb{E}[X^n]$ plus terms with θ.
$\big\vert_{\theta=0}$	evaluate at $\theta = 0$, which removes the higher terms.

1.7 The normal distribution: PDF [S11]

Gaussian integral:

$$I = \int_{-\infty}^{\infty} e^{-y^2}\,\mathrm{d}y, \qquad I^2 = \pi \;\Rightarrow\; I = \sqrt{\pi}.$$

where

I	the Gaussian integral (no elementary antiderivative).
I^2	write as a double integral $\iint e^{-(x^2+y^2)}\mathrm{d}x\,\mathrm{d}y$ and switch to polar coordinates to get π *[Beyond slides]*.

Substituting $y = x/\sqrt{2}$, then $y = (x-\mu)/(\sqrt{2}\sigma)$:

$$\int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}}_{\phi(x)}\,\mathrm{d}x = 1, \qquad \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}}_{f(x)}\,\mathrm{d}x = 1.$$

where

$\phi(x)$	standard normal PDF ($\mu = 0$, $\sigma = 1$).
$f(x)$	normal PDF with mean μ and variance σ^2.
$\frac{1}{\sqrt{2\pi}\,\sigma}$	normalising constant that makes the total area 1 (the $\sqrt{\pi}$ from I, $\sqrt{2}\sigma$ from the substitution).
$(x-\mu)^2/(2\sigma^2)$	squared distance from the mean in units of the variance; controls the bell shape.

1.8 MGF of the normal distribution [S12–S13]

For $X \sim N(\mu, \sigma^2)$:

$$M_X(\theta) = \mathbb{E}\big[e^{\theta X}\big] = \exp\Big(\mu\theta + \tfrac{1}{2}\sigma^2\theta^2\Big), \qquad M_Z(\theta) = \exp\Big(\tfrac{\theta^2}{2}\Big) \text{ for } Z \sim N(0,1).$$

where

$\mu\theta$	contribution of the mean.
$\frac{1}{2}\sigma^2\theta^2$	contribution of the variance (the convexity term).
Z	standard normal ($\mu = 0$, $\sigma = 1$).

Derivation by completing the square [S12–S13], key steps:

$$\mathbb{E}\big[e^{\theta X}\big] = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} \exp\Big[-\frac{\big(x-(\mu+\sigma^2\theta)\big)^2}{2\sigma^2} + \mu\theta + \frac{\sigma^2\theta^2}{2}\Big]\mathrm{d}x = e^{\mu\theta+\frac{\sigma^2\theta^2}{2}} \underbrace{\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{-\frac{[x-(\mu+\sigma^2\theta)]^2}{2\sigma^2}}\mathrm{d}x}_{=1}.$$

where

$\theta x - \dfrac{(x-\mu)^2}{2\sigma^2}$	combine the exponents of $e^{\theta x}$ and the normal PDF over a common denominator $2\sigma^2$.
$\mu + \sigma^2\theta$	the centre of the completed square: the exponent becomes a normal density shifted to mean $\mu + \sigma^2\theta$.
$\mu\theta + \dfrac{\sigma^2\theta^2}{2}$	the leftover constant after completing the square; it comes out of the integral.
$= 1$	the remaining integral is a normal PDF with mean $\mu + \sigma^2\theta$, variance σ^2, so it integrates to 1.

[Beyond slides] Check: $M'(0) = \mu$, $M''(0) = \mu^2 + \sigma^2$, so $V[X] = \sigma^2$ ✓.

1.9 Lognormal random variables [S14]

$$X \sim N(\mu, \sigma^2), \qquad Y = e^X, \qquad \ln Y \sim N(\mu, \sigma^2), \qquad \mathbb{E}[Y] = \mathbb{E}[e^X] = M_X(1) = \exp\Big(\mu + \tfrac{1}{2}\sigma^2\Big).$$

where

X	normal variable (e.g. a continuously compounded return).
$Y = e^X$	lognormal variable: always positive (e.g. a price $S_T = S_0 e^X$).
$\ln Y$	its log is normal, hence the name.
$M_X(1)$	MGF evaluated at $\theta = 1$, since $e^X = e^{1\cdot X}$.
$\frac{1}{2}\sigma^2$	convexity term: $\mathbb{E}[e^X] > e^{\mathbb{E}[X]}$ (Jensen's inequality, *[Beyond slides]*).

[Beyond slides] Also $V[Y] = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$ and $\mathbb{E}[Y^n] = M_X(n) = e^{n\mu+\frac{1}{2}n^2\sigma^2}$.

Slide question [S14]: why normal for returns, lognormal for prices? *[Beyond slides]*: returns can be negative and, over many short periods, add up (CLT ⇒ normal); prices cannot go below zero and compound multiplicatively, so $S_T = S_0 e^{\text{sum of returns}}$ is lognormal; limited liability is respected.

1.10 Two-state model: hedging and the no-arbitrage price [S15–S17]

Setup [S15]: $S_0 = 100$, tomorrow $S_1 = 110$ or 90; ATM call $K = 100$ pays $(S_1 - K)^+ = 10$ or 0; $r = 0$. Price must lie in $[0, 10]$.

Hedge [S16]: sell one call, buy Δ shares; choose Δ so the portfolio is worth the same in both states:

$$\underbrace{\Delta \times 110 - 1 \times 10}_{\text{up}} = \underbrace{\Delta \times 90 - 1 \times 0}_{\text{down}} \quad \Rightarrow \quad \Delta = \frac{1}{2}.$$

where

Δ	number of shares bought per call sold (hedge ratio).
$\Delta \times 110,\ \Delta \times 90$	value of the shares in the up / down state.
$-1 \times 10,\ -1 \times 0$	the call we owe in each state (short position).
$\Delta = \frac{1}{2}$	solves the equation; generally $\Delta = \frac{C_u - C_d}{S_u - S_d} = \frac{10-0}{110-90}$ *[Beyond slides]*.

The portfolio is worth $110 \cdot \frac{1}{2} - 10 = 90 \cdot \frac{1}{2} = 45$ in both states, so it is riskless; with $r = 0$ it must cost 45 today:

$$100 \times \tfrac{1}{2} - C_0 = 45 \quad \Rightarrow \quad C_0 = 5.$$

where

$100 \times \frac{1}{2}$	cost of the Δ shares today.
$-C_0$	cash received for selling the call today.
45	today's value of a riskless payoff of 45 when $r = 0$ (otherwise $45/(1+r)$).
C_0	the no-arbitrage call price: any other price lets someone lock in a riskless profit.

No probabilities were used [S17]. They only matter through which states are possible: an up probability of 1 is impossible (borrow at r, buy stock, sure profit), and likewise a down probability of 1 (short the stock, invest the cash). Both states must have non-zero probability.

1.11 Two-state model: risk-neutral probabilities [S18–S19]

Risk-neutral probability [S18]: choose the up probability p^* that matches the no-arbitrage price.

$$\begin{cases} \mathbb{E}[S_1] = 110\,p^* + 90\,(1-p^*) \\ \mathbb{E}[C_1] = 10\,p^* \end{cases} \quad \Rightarrow \quad \mathbb{E}[C_1] = C_0 = 5 \iff p^* = \tfrac{1}{2}, \quad \text{and then} \quad \mathbb{E}[S_1] = 100 = S_0.$$

where

p^*	risk-neutral probability of the up state (a pricing device, not a forecast).
$1 - p^*$	risk-neutral probability of the down state.
$\mathbb{E}[S_1]$	expected stock price tomorrow under p^*.
$\mathbb{E}[C_1]$	expected call payoff under p^* (the down payoff is 0).
$\mathbb{E}[S_1] = S_0$	with $r = 0$, under p^* the stock has zero expected gain: the defining property of the risk-neutral measure (with $r \neq 0$: $\mathbb{E}[S_1] = S_0(1+r)$, Session 2).

- *[Beyond slides]* General one-step formula: $p^* = \dfrac{S_0(1+r) - S_d}{S_u - S_d}$, $C_0 = \dfrac{p^* C_u + (1-p^*)C_d}{1+r}$.
- **Why it works** [S19]: the no-arbitrage price uses only which probabilities are zero or non-zero, so an arbitrage either exists for all $0 < p < 1$ or for none.
- In the real world one expects $p \geq \frac{1}{2}$ (risk aversion: investors need a premium), so expected stock and option values would exceed the no-arbitrage prices. Under p^* no premium is needed: **the ability to hedge has removed the risk premium** [S19].

1.12 Three-state (trinomial) model and incompleteness [S20–S22]

Setup [S20]: add a middle state: $S_1 \in \{110, 100, 90\}$, call payoffs $\{10, 0, 0\}$.

Hedge attempt [S21]: portfolio "Δ shares, short 1 call" is worth

$$110\Delta - 10, \qquad 100\Delta, \qquad 90\Delta.$$

where

$110\Delta - 10$	up state: shares minus the call payoff owed.
$100\Delta,\ 90\Delta$	middle and down states: the call expires worthless.

One unknown (Δ) but two constraints (up = middle and middle = down), so no Δ makes it riskless:

- Match middle and down: $\Delta = 0$, i.e. no hedge at all.
- Match up and down: $\Delta = \frac{1}{2}$, portfolio worth 45, 50, 45. It is worth at least 45 and sometimes more, so it must cost more than 45 today:

$$50 - C_0 > 45 \quad \Rightarrow \quad C_0 < 5.$$

where

50	cost of $\frac{1}{2}$ share today ($\frac{1}{2} \times 100$).
$-C_0$	premium received for the call.
> 45	a portfolio paying ≥ 45, and > 45 with positive probability, must cost > 45 (else arbitrage).
$C_0 < 5$	an *upper bound* only, not a unique price. *[Beyond slides]* Lower bound: the call pays ≥ 0 and > 0 in one state, so $C_0 > 0$.

Risk-neutral approach [S22]: need $\mathbb{E}[S_1] = 100$, so $\mathbb{P}^*(110) = \mathbb{P}^*(90) = p^*$ and $\mathbb{P}^*(100) = 1 - 2p^*$:

$$\mathbb{E}[S_1] = 110p^* + 100(1-2p^*) + 90p^* = 100 \;\; \forall\, p^*, \qquad C_0 = \mathbb{E}[C_1] = 10\,p^*, \qquad 0 < p^* < \tfrac{1}{2} \;\Rightarrow\; 0 < C_0 < 5.$$

where

p^*	risk-neutral probability of each of the up and down states (they must be equal to keep the mean at 100).
$1-2p^*$	risk-neutral probability of the middle state; must be positive, hence $p^* < \frac{1}{2}$.
$0 < p^*$	every state must stay possible (equivalence).
$10p^*$	expected call payoff (only the up state pays).
$(0, 5)$	the set of arbitrage-free prices: an interval, matching the hedging bounds.

Incomplete market [S22]: portfolios of stock and cash cannot replicate every payoff, so prices are only pinned down to an interval, not a unique value. In the two-state world the upper and lower bounds coincide and the price is unique [S20].

1.13 Implementation *[Beyond slides]* (tested)

```
import numpy as np
mu, s = 0.1, 0.2
M = lambda th: np.exp(mu*th + 0.5*s**2*th**2)       # normal MGF
h = 1e-3
m1 = (M(h) - M(-h)) / (2*h)                          # E[X]   = M'(0)
m2 = (M(h) - 2*M(0) + M(-h)) / h**2                  # E[X^2] = M''(0)
print(round(m1, 6), round(m2, 6), round(mu**2 + s**2, 6))   # 0.1 0.05 0.05
# Two-state hedge: S0=100 -> 110/90, ATM call, r=0
Su, Sd, Cu, Cd, S0 = 110, 90, 10, 0, 100
delta = (Cu - Cd) / (Su - Sd)
C0 = delta*S0 - (delta*Su - Cu)                      # cost of shares minus sure portfolio value
p = (S0 - Sd) / (Su - Sd)
print(delta, C0, p*Cu + (1-p)*Cd)                    # 0.5 5.0 5.0
# Trinomial: call price 10 p* as p* sweeps (0, 1/2)
print(np.round(10*np.linspace(1e-6, 0.5-1e-6, 5), 4))   # [0. 1.25 2.5 3.75 5.]
```

2 Comparison tables

2.1 Raw vs central vs standardised moments

	Raw $\mathbb{E}[X^k]$	Central $\nu_k = \mathbb{E}[(X-\mu)^k]$	Standardised $\gamma_k = \nu_k/\sigma^k$
About	origin $x = 0$	mean μ	mean, in units of σ
$k = 1$	mean μ	0	0
$k = 2$	$\mu^2 + \sigma^2$	variance σ^2	1
$k = 3$	—	ν_3	skewness (0 for normal)
$k = 4$	—	ν_4 ($3\sigma^4$ for normal)	kurtosis (3 for normal)
Units	units^k	units^k	dimensionless

2.2 Normal vs lognormal

	Normal $X \sim N(\mu, \sigma^2)$	Lognormal $Y = e^X$
Support	$(-\infty, \infty)$	$(0, \infty)$
Mean	μ	$e^{\mu + \frac{1}{2}\sigma^2}$
Variance	σ^2	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$ *[Beyond slides]*
Skew	symmetric	right-skewed
MGF / moments	$e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$	$\mathbb{E}[Y^n] = M_X(n)$
Used for	returns	prices

2.3 Two-state vs three-state model

	Two-state [S15–S19]	Three-state [S20–S22]
States / instruments	2 states, 2 instruments	3 states, 2 instruments
Hedge	$\Delta = \frac{1}{2}$, riskless	no riskless Δ
Risk-neutral p^*	unique: $\frac{1}{2}$	any $p^* \in (0, \frac{1}{2})$
Call price	unique: 5	interval: $(0, 5)$
Market	complete	incomplete

2.4 Measures of the centre

Mean	$\int x f(x)\,\mathrm{d}x$; sensitive to outliers (80.68° in [S5])
Median	$F(M) = \frac{1}{2}$; robust (80° in [S5])
Mode	$\arg\max f(x)$; may not be unique (80° in [S5])

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Expectation and variance [S2–S3]

- $\mathbb{E}[g(X)] = \int g(x)f(x)\,\mathrm{d}x$; $\mathbb{E}[a] = a$, $\mathbb{E}[ag(X)] = a\mathbb{E}[g(X)]$, $\mathbb{E}[s+t] = \mathbb{E}[s] + \mathbb{E}[t]$.
  where f = PDF; g = any function; a = constant; s, t = functions of X.
- $V[X] = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2$; $V[aX+b] = a^2V[X]$.
  where $\mu = \mathbb{E}[X]$; a, b = constants.

Location [S4]

- $F(Q_{\text{lower}}) = \frac{1}{4}$, $F(M) = \frac{1}{2}$, $F(Q_{\text{upper}}) = \frac{3}{4}$, $F(P_n) = \frac{n}{100}$.
  where F = CDF; M = median; Q = quartiles; P_n = n-th percentile.

Moments [S6–S9]

- $\nu_k = \mathbb{E}[(X-\mu)^k]$; $\nu_3 = \mathbb{E}[X^3] - 3\mathbb{E}[X]\mathbb{E}[X^2] + 2\mathbb{E}[X]^3$.
  where ν_k = k-th central moment; $\mathbb{E}[X^k]$ = raw moments.
- $\gamma_k = \nu_k/\sigma^k$; skewness γ_3, kurtosis γ_4; normal: $\gamma_3 = 0$, $\gamma_4 = 3$; excess $\gamma_4 - 3$.
  where σ = standard deviation.

MGF, normal, lognormal [S10–S14]

- $M_X(\theta) = \mathbb{E}[e^{\theta X}]$; $\mathbb{E}[X^n] = M_X^{(n)}(0)$.
  where θ = dummy variable; $M^{(n)}$ = n-th derivative.
- $X \sim N(\mu, \sigma^2)$: $M_X(\theta) = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$; $M_Z(\theta) = e^{\theta^2/2}$.
  where $Z \sim N(0,1)$.
- $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$; $\int e^{-y^2}\mathrm{d}y = \sqrt{\pi}$.
  where f = normal PDF.
- $Y = e^X$: $\mathbb{E}[Y] = e^{\mu + \frac{1}{2}\sigma^2}$.
  where $X \sim N(\mu, \sigma^2)$; $\frac{1}{2}\sigma^2$ = convexity term.

Lattice pricing [S15–S22]

- $\Delta = \frac{C_u - C_d}{S_u - S_d}$; $C_0 = \Delta S_0 - \frac{\Delta S_u - C_u}{1+r}$ *[Beyond slides].*
  where C_u, C_d = option payoffs; S_u, S_d = stock prices up/down; r = rate (0 on the slides).

- $p^* = \frac{S_0(1+r)-S_d}{S_u-S_d}$; $C_0 = \frac{p^*C_u+(1-p^*)C_d}{1+r}$ (slides: $r=0$, $p^*=\frac{1}{2}$, $C_0=5$).
  where p^* = risk-neutral up probability.
- Trinomial: $p^* \in (0, \frac{1}{2})$, middle $1-2p^*$, $C_0 = 10p^* \in (0,5)$.
  where equal up/down probabilities keep $\mathbb{E}[S_1] = S_0$.

4 Glossary

- **Arbitrage:** a riskless profit with no initial cost; prices must rule it out [S16].
- **Arbitrage-free price:** a price that admits no arbitrage [S20].
- **Central moment ν_k:** $\mathbb{E}[(X-\mu)^k]$ [S7].
- **CDF $F(x)$:** $\mathbb{P}(X \le x)$ [S4].
- **Complete market:** every payoff can be replicated (two-state model).
- **Excess kurtosis:** $\gamma_4 - 3$ [S9].
- **Expectation:** probability-weighted average $\int gf$ [S2].
- **Hedge ratio Δ:** shares held per option to remove risk [S16].
- **Incomplete market:** not every payoff can be replicated; prices lie in an interval [S22].
- **Kurtosis γ_4:** standardised fourth central moment [S9].
- **Lattice (tree) model:** discrete states for future prices [S15].
- **Lognormal:** Y with $\ln Y$ normal [S14].
- **Median, mode, quartile, percentile:** [S4].
- **Moment:** $\mathbb{E}[X^k]$ [S6].
- **MGF $M_X(\theta)$:** $\mathbb{E}[e^{\theta X}]$ [S10].
- **PDF $f(x)$:** probability density function [S2].
- **Replication:** building a payoff from stock and cash [S22].
- **Risk-neutral probability p^*:** probability that reproduces arbitrage-free prices; stock earns the risk-free rate [S18].
- **Risk premium:** extra expected return demanded for bearing risk; removed by hedging [S19].
- **Skewness γ_3:** standardised third central moment [S9].
- **Variance, standard deviation:** $\mathbb{E}[(X-\mu)^2]$, σ [S3].

5 Self-test

Questions

Q1 ([S3]). Show $V[aX+b] = a^2V[X]$ and $V[X] = \mathbb{E}[X^2] - \mu^2$.

Q2 ([S5]). For the 110-day temperature data, find the mean, median and mode.

Q3 ([S8]). Express ν_3 in terms of raw moments.

Q4 ([S9]). What do $\gamma_3 < 0$ and $\gamma_4 - 3 > 0$ say about a return distribution?

Q5 ([S10]). Show $\mathbb{E}[X^n] = M_X^{(n)}(0)$. Use it to get the mean and variance of $N(\mu, \sigma^2)$.

Q6 ([S12–S13]). Derive $M_X(\theta)$ for $X \sim N(\mu, \sigma^2)$ by completing the square.

Q7 ([S14]). For $X \sim N(\mu, \sigma^2)$, find $\mathbb{E}[e^X]$ and $\mathbb{E}[e^{2X}]$. Why model prices as lognormal?

Q8 ([S15–S16]). $S_0 = 100$, $S_1 \in \{110, 90\}$, $r = 0$. Price the ATM call by hedging. Repeat for an ATM put.

Q9 ([S17–S19]). Find p^* and verify both prices of Q8. Why is $p = 1$ impossible?

Q10 ([S20–S22]). In the trinomial model, find the range of arbitrage-free call prices by hedging and by risk-neutral probabilities. Why is the market incomplete?

Solutions

S1. $\mathbb{E}[aX+b] = a\mu+b$, so $V[aX+b] = \mathbb{E}[(aX-a\mu)^2] = a^2V[X]$. $\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2$.

S2. Mean $8875/110 \approx 80.68$; median: 55th and 56th values are 80, so 80; mode 80.

S3. $\nu_3 = \mathbb{E}[X^3] - 3\mu\mathbb{E}[X^2] + 3\mu^2\mathbb{E}[X] - \mu^3 = \mathbb{E}[X^3] - 3\mathbb{E}[X]\mathbb{E}[X^2] + 2\mathbb{E}[X]^3$.

S4. Negative skew: long left tail (large losses more likely than equally large gains). Positive excess kurtosis: fatter tails and sharper peak than a normal with the same mean and variance.

S5. Differentiate the series $\sum \mathbb{E}[X^k]\theta^k/k!$ n times and set $\theta = 0$. $M'(\theta) = (\mu + \sigma^2\theta)M$, so $M'(0) = \mu$; $M''(0) = \sigma^2 + \mu^2$; variance σ^2.

S6. Exponent $\theta x - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{[x-(\mu+\sigma^2\theta)]^2}{2\sigma^2} + \mu\theta + \frac{1}{2}\sigma^2\theta^2$; the remaining integral is a normal PDF $= 1$.

S7. $M_X(1) = e^{\mu+\frac{1}{2}\sigma^2}$; $M_X(2) = e^{2\mu+2\sigma^2}$. Prices are positive and compound multiplicatively; returns add.

S8. Call: $\Delta = \frac{10-0}{20} = \frac{1}{2}$, portfolio 45, $C_0 = 50 - 45 = 5$. Put payoffs $0, 10$: buy Δ shares with the long put: $110\Delta = 90\Delta + 10 \Rightarrow \Delta = \frac{1}{2}$, portfolio 55, $P_0 = 55 - 50 = 5$. (Put–call parity with $r = 0$: $C - P = S_0 - K = 0$ ✓.)

S9. $p^* = \frac{100-90}{20} = \frac{1}{2}$: $C_0 = 10 \cdot \frac{1}{2} = 5$, $P_0 = 10 \cdot \frac{1}{2} = 5$. If the stock surely rises, borrow and buy: riskless profit. So both states must be possible.

S10. Hedging: $\Delta = \frac{1}{2}$ gives $45, 50, 45 \Rightarrow C_0 < 5$; payoff ≥ 0 and sometimes $> 0 \Rightarrow C_0 > 0$. Risk-neutral: $p^* \in (0, \frac{1}{2})$, $C_0 = 10p^* \in (0, 5)$. One instrument (Δ) cannot match three states, so payoffs are not replicable.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	PDF; $\mathbb{E}[g(X)]$; expectation properties; $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$	1.1
S3	Mean and variance; variance properties; standard deviation	1.2
S4	Mode, median, quartiles, percentiles	1.3
S5	Example: median temperature	1.3
S6	Moments; $V[X] = \mathbb{E}[X^2] - \mu^2$	1.4
S7	Central moments ν_k	1.4
S8	Example: ν_3 in terms of moments	1.4
S9	Standardised moments; skewness; kurtosis; excess kurtosis	1.5
S10	Moment generating function; moments from derivatives	1.6
S11	Gaussian integral; normal PDF	1.7
S12	Normal MGF: completing the square (start)	1.8
S13	Normal MGF (end); standard normal MGF	1.8
S14	Lognormal variables; $\mathbb{E}[e^X]$; returns vs prices	1.9
S15	Two-state model setup	1.10
S16	Hedging; $\Delta = \frac{1}{2}$; $C_0 = 5$	1.10
S17	Role of probabilities; no-arbitrage	1.10
S18	Risk-neutral probability $p^* = \frac{1}{2}$	1.11
S19	Real-world vs risk-neutral; risk premium removed	1.11
S20	Three-state model setup	1.12
S21	Hedging fails; upper bound $C_0 < 5$	1.12
S22	Risk-neutral range; incomplete market	1.12

Appendix B. Flag summary (△!)

1. [S3] "Variance is always positive": it is non-negative; $V[a] = 0$.
2. [S9] v_4 should read ν_4.
3. [S2] $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ does hold under independence; the slide's "in general" is the key qualifier.
4. [S21] The hedging argument only gives an upper bound; the lower bound $C_0 > 0$ comes from the payoff being non-negative.
5. [S5, S8, S14, S17, S21] Questions posed without full answers (median, "why sensible", "why?", "which two?"); the answers here are my own.